

CARDIOVASCULAR MEDS

BETA BLOCKERS (SUFFIX: -olol)

ACTION: Inhibits beta receptors from doing their job → lower HR/BP

KEY: Beta₂ receptors work on the lungs so we may see bronchospasm → **DO NOT** give to asthma/COPD patients

CONSIDERATIONS:

- can be given to ♥ failure pts
- may mask sx of hypoglycemia
- Always check HR/BP BEFORE giving → **HOLD** if BP < 90 / HR < 60
- Watch orthostatic hypotension, wheezing, ↓ HR, ↓ BP

SELECTIVE (♥ only)

- Atenolol
- Metoprolol

NONSELECTIVE:

- Propranolol
- Sotalol

ACE INHIBITORS (-pril) → 1st choice ARB's (-sartan) → 2nd choice

ACTION: **BOTH** inhibit the normal function of the RAAS system → lower BP by getting rid of Na⁺/H₂O and ↑ K⁺

KEY: Teratogenic!

- > With ACE → cough/angioedema (Risk: Airway)
- > ↑ of K⁺ in the body!

CONSIDERATIONS:

- K⁺ levels → EKG (Peak T wave / ST elevation)
- K⁺ diet → avoid K⁺ rich foods
- Watch BP / orthostatic hypotension

CALCIUM CHANNEL BLOCKERS

ACTION: Blocks movement of calcium which relaxes smooth muscle → ↓ resistance → ↓ BP

TYPES: - DILPINE (only lowers BP)
- ZEM / -AMIL (lowers both HR/BP)

CONSIDERATIONS:

- Do NOT give to ♥ failure pt (will worsen)
- Always check HR/BP BEFORE giving → **HOLD** if BP < 90 / HR < 60
- Watch orthostatic hypotension / **TOO LOW** HR/BP
- Headache = **NORMAL** finding

DIGOXIN (for HEART FAILURE)

→ can also be given as an antiarrhythmic

ACTION: DOES NOT affect BP, **ONLY HR**
"DIG's for a forceful contraction"

THERAPEUTIC RANGE: 0.5 - 2.0

CONSIDERATIONS:

- Does **NOT** waste potassium → don't change K⁺ diet (↓ K⁺ = ↑ Risk toxicity)
- Monitor** kidney function (BUN, Cr)
(↓ kidney function = ↑ risk toxicity)
- S/S Toxicity:** vision change, dizzy, N/V, irregular pulse, confusion

VASODILATORS (can be used in heart failure)

ACTION: Relaxes smooth muscle = ↓ BP, ↑ O₂ to major organs
NORMAL S/S: headache, flushing, hypotension

CONSIDERATIONS:

- Do **NOT** take with Viagra (sildenafil)
- HOLD** if BP systolic < 100 or drops > 30 mmHg
- Monitor BP + orthostatic hypotension

EX: Nitroglycerin, Hydralazine, Nitroprusside, Isosorbide

KEY TAKEAWAY:

- Lowers BP: All except Digoxin
- Lowers HR: Beta blockers, CCB's (zem and amil), Digoxin
- Do NOT give with ♥ failure: CCB's
- YES for ♥ failure: Digoxin, BB's, vasodilators, diuretics
- NOT for pregnancy: ACE / ARB's
- Watch vitals! **Don't stop abruptly!**

DIURETICS

① POTASSIUM WASTING (-IDE)

ACTION: DRY up the body by excreting (wasting) potassium → increase urination

KEY: ONLY give if K^+ level is in range → 3.5-5.0 mEq/L

USE: Heart failure, worsening crackles, new edema in the legs, rapid weight gain

A.) LOOP DIURETICS: Furosemide
Bumetanide
Torsemide

B.) THIAZIDE DIURETICS: Hydrochlorothiazide
Chlorothiazide

KEY: Watch for **decreased K^+** = decreased heart beats
(flat T wave, ST depression, U wave)

• **Encourage K^+ rich foods** → Bananas, green leafy veggies, melon
(No licorice Root = ↓ K^+)

• **Furosemide:** given **too fast** = tinnitus, hypotension
given **too long** = kidney damage, ↑ BUN, ↑ Cr, ↓ K^+

② POTASSIUM SPARING (SPIRONOLACTONE)

ACTION: Blocks Aldosterone directly = release Na^+ and H_2O
"saves" potassium in the body

KEY: Watch for **increased K^+** = increased heart beats
(Peak T wave, ST elevation)

• **Avoid K^+ rich foods** / NO salt substitutes

• Make sure to put on a **cardiac monitor**

③ MANNITOL (OSMOTIC DIURETIC)

ACTION: ↓ thickness of blood → ↑ cerebral blood flow → ↓ blood volume / IOP

USE: Treat increased intracranial pressure; lower eye pressure

SIDE EFFECTS: Headache, Nausea, acute renal failure (large doses)

MONITOR: Neurologic and renal status

KEY: FOR POTASSIUM WASTING + SPARING:

• Always check BP; **HOLD** if systolic < 100

• Always check BUN, Cr, K^+ levels; Heart monitor!

• Give in the **morning**, not at night!

• **DAILY WEIGHTS**

• **No sodium meds**
(cough/flu, antacids, NSAIDs)

• **Teach** orthostatic hypo,
Risk sunburns

RESPIRATORY MEDS

① BETA₂ AGONISTS (SUFFIX: -terol)

ACTION: Enhance the effects of Beta₂ in the body = **BRONCHODILATION**

TYPES: SHORT ACTING

➤ For acute attack

➤ **NOT** used for daily tx (>2x/WK = BAD)

➤ Can be used with steroid or anticholinergic

EX: ALBUTEROL

LONG ACTING

➤ **NOT** for acute attack

➤ should **ONLY** be taken in combo with a corticosteroid (never alone)

EX: SALMETEROL

SIDE EFFECTS: tachycardia, jittery, nervous, palpitations, tremor

KEY: If used in combo with steroid or anticholinergic → **GIVE FIRST**

➤ These are all typically **inhaled**

② CORTICOSTEROIDS (SUFFIX: -sone, -ide)

USED FOR: Asthma, rhinitis, autoimmune disorders

ACTION: Reduce the inflammatory response by suppressing the immune system

➤ can be used **alone** or with a Beta₂ agonist

SIDE EFFECTS: osteoporosis (affects calcium)

↑ glucose, ↓ K⁺, fluid retention + urinary retention

PUD, **IMMUNOSUPPRESSION!**

MAY CAUSE: ➤ headache

➤ sore throat

➤ hoarseness

➤ candidiasis



ORAL THRUSH

➤ use a spacer & rinse mouth!

Remember:

➤ ↑ stress = may need ↑ dose

➤ Never stop abruptly = **TAPER**

➤ monitor s/s of infection

➤ **AVOID NSAIDS**

EX: Beclomethasone, Fluticasone, Budesonide, Prednisone

RESPIRATORY MEDS

③ INHALED ANTICHOLINERGICS (SUFFIX: -ropium)

USED FOR: COPD, bronchitis, emphysema, asthma, allergies

ACTION: Block the parasympathetic nerve reflexes that cause the airway to constrict = **BRONCHODILATION**

• short acting: IPRATROPIUM

• Long acting: TIOTROPIUM

can be used with
albuterol = DUONEB

SIDE EFFECTS: Dry mouth, throat irritation
hoarseness, bitter taste

KEY: give sugarless
candy or ↑ fluids

④ LEUKOTRIENE MODIFIERS (SUFFIX: -lukast)

USED FOR: Asthma, exercise induced bronchoconstriction

ACTION: Blocks the normal function of leukotriene to dilate the smooth muscle and ↓ mucus

KEY: Leukotriene typically causes smooth muscle to constrict and increase mucus production → **we are stopping this**

SIDE EFFECTS: headache, sore throat, cold symptoms

KEY: **NOT** for acute attack!

MONTELUKAST: Take in the evening or 2hr before exercise

ZAFIRLUKAST: Take on empty stomach, check liver enzymes

⑤ XANTHINES (THEOPHYLLINE)

USED FOR: Long term asthma / COPD

ACTION: ↑ cAMP = **BRONCHODILATION**

SIDE EFFECTS: **MINOR**

- GI upset
- Headache
- nervous

MAJOR

- seizure
- dysrhythmias

KEY: Avoid caffeine!

Not as common toxicity
level: 10-20mcg/mL

RESPIRATORY MEDS

EXPECTORANTS

USED FOR: Nonproductive cough

ACTION: Decrease viscosity of lung secretions; this will make the cough more productive

SIDE EFFECTS: Headache, dizzy, N/V

➤ **Prolonged use:** may mask an underlying serious condition for which you have the cough

EX: GUAIFENESIN

MUCOLYTICS

USED FOR: Pulmonary disorders with thick mucus (CF, COPD, PNA)

ACTION: Breaks down tenacity and viscosity of secretions

SIDE EFFECTS: Bronchospasm (watch w/ asthma)
➤ N/V, Rash
➤ Headache, dizzy

WATCH: Pts with PUD, asthma, esophageal varices

EX: ACETYLCYSTEINE
DORNASE ALFA

DECONGESTANTS (ORAL+TOPICAL)

USED FOR: Rhinitis, sinusitis, common cold

ACTION: Vasoconstricts the respiratory mucosa = ↓ inflammation

Relief of pain and congestion

SIDE EFFECTS: Palpitations, restless
➤ Weakness, HTN, nervous
{ REBOUND CONGESTION }

WATCH: If taking with other OTC products

EX: PSEUDOEPHEDRINE
PHENYLEPHRINE

ANTIHISTAMINES

USED FOR: Seasonal allergies, urticaria, angioedema, allergic conjunctivitis

ACTION: Blocks histamine at Histamine-1 receptor sites

➤ Anticholinergic + antipyretic effect

SIDE EFFECTS: itching, sneezing, hives, GI upset, nausea, dry mouth, constipation, sedation, drowsy
Σ FALL RISK

CONTRAINDICATED: PREGGO }

EX: 1st GEN: DIPHENHYDRAMINE
2ND GEN: LORATADINE / CETIRIZINE

NONSTEROIDAL ANTIALLERGY GROMOLYN

USED FOR: 3rd line drug for mastocytosis

ACTION: stops mast cells from secreting histamine

SIDE EFFECTS: Burning in nose
itchy, watery eyes
Bad taste in mouth

Σ **KEY:** NOT QUICK = LONG TERM!

SUBCUTANEOUS ANTI-INFLAMMATORY OMALIZUMAB

USED FOR: Allergic asthma, chronic idiopathic urticaria

ACTION: Blocks the role of IgE = ↓ allergic response

SIDE EFFECTS: itching, mild rash
➤ Nausea, dizzy, ear pain
➤ cold symptoms
➤ pain in arms or legs

Σ **KEY:** NOT QUICK
NO LIVE VACCINES d/t immunosupp.

HYPERLIPIDEMIC MEDS

KEY: For all medications in this class:

1. All will affect the liver and lipid profile

2. Medications will most likely **NOT** be prescribed to lower cholesterol during pregnancy.

3. **LFT'S:** AST: 10-40 U/L

ALT: 7-56 U/L

ALP: 40-120 U/L

Bili: <1 mg/dL

LIPID PANEL: Total cholesterol: <200

Triglycerides: <150

LDL: <100

HDL: >60

① STATINS (SUFFIX: -statin)

USED FOR: Decrease cholesterol + Reduce risk of MI or stroke

ACTION: Blocks the enzyme HMG-CoA reductase to slow down cholesterol production

SIDE EFFECTS: Liver toxicity

• myopathy (muscle pain)

• rhabdomyolysis

WATCH: LFT's and CK levels (Nml: 22-198 U/L) → metabolized in liver
excreted by kidneys

• Avoid Alcohol

Σ • TAKE IN THE EVENING

② BILE ACID SEQUESTRANTS (PREFIX: cole-)

AKA: RESINS

USED FOR: lowering cholesterol (LDL)

ACTION: ↓ LDL by binding bile acids in the gut = pee out = pulls more cholesterol from bloodstream to make bile

SIDE EFFECTS: constipation, flatulence
GI disturbance, bad taste

KEY: • Teach ↑ fiber / ↑ fluids = interferes with absorption of fat soluble vitamins A, D, E, K (a fat DEK of cards)

• TAKE WITH FOOD

EX: COLESEVELAM
COLESTPAL

HYPERLIPIDEMIC MEDS

③ CHOLESTEROL ABSORPTION INHIBITORS (EZETIMIBE:)

"zip" through the intestines

USED FOR: lowering cholesterol (LDL)

ACTION: ↓ LDL by inhibiting absorption of cholesterol in the intestines

SIDE EFFECTS: Liver toxicity

› Myopathy (muscle pain) → rhabdomyolysis

› May cause headache, dizziness, fatigue

WATCH: LFT's + CK levels (Nml: 22-198 U/L)

› Watch in pregnancy IF given

④ NIAICIN/NICOTINIC ACID (vit. B3)

USED FOR: lowering cholesterol (LDL) and Triglycerides
↑ HDL ü

ACTION: when given well above the vitamin requirement it will reduce the release of fatty acids

SIDE EFFECTS: Liver toxicity

› Myopathy, gout (hyperuricemia)

› N/V/D, hyperglycemia

Σ › Flushing → USE ASPIRIN

WATCH: LFT's + CK levels (Nml: 22-198 U/L)

› Blood glucose

⑤ FIBRATES (All include "fib" in the name) →

Gemfibrozil
Clofibrate
Fenofibrate

USED FOR: lowering triglyceride levels

↑ HDL ü

ACTION: Lipoprotein lipolysis

SIDE EFFECTS: Liver toxicity

Myopathy

Gallstones, GI upset, N/V

WATCH: LFT's + CK levels (Nml: 22-198 U/L)

KEY: Give 30min before breakfast + dinner!

ALL ABOUT INSULIN

RAPID-ACTING	SHORT-ACTING	INTERMEDIATE-ACTING	LONG-ACTING
1. Aspart THINK: "Move your Ass" → Ass-part 2. Lispro THINK: "Let's go!!" → Lispro 3. Glulisine THINK: Glue dries fast Onset: 15 MIN! Peak: 30-90 minutes Duration: 3-5 hours	AKA: Regular Insulin KEY: This is the ONLY insulin type given IV route - Can be given with NPH at the same time in the same syringe - Can be given with long-acting at the same time in a different syringe Onset: 30-60 minutes Peak: 2-4 hours Duration: 5-8 hours	AKA: NPH KEY: If given with regular insulin, draw up: clear-to-cloudy THINK: R-N → Regular before NPH (clear before cloudy) Given 2x/day Onset: 60-120 minutes Peak: 4-12 hours Duration: 14 hours (hence, given 2x/day)	KEY: NO PEAK CAN'T BE MIXED WITH OTHER INSULIN! 1. Detrimir THINK: "Lasts all year" → lasts a long time 2. Lantus THINK: "Lantern" → lanterns burn for a long time 3. Glargine THINK: "Large" → lasts for a large amount of time Onset: 60-120 minutes Peak: NO PEAK Duration: 24 hours

WHEN DO YOU EAT?

- Rapid-acting:** Covers insulin needs for meals eaten at the **same** time of injection
- Short-acting (Regular):** Covers insulin needs for meals eaten within **30-60 minutes** of injection
- Intermediate-acting (NPH):** Covers insulin needs for **half** the day or overnight; typically given morning and night
- Long-acting:** Covers insulin needs for the **full** day; can be combined with other insulin but never mixed

RULES OF INSULIN

- Watch for signs and symptoms of hypoglycemia → shaky, clammy, pale, sweaty
 - THINK: "Cool and clammy, give me candy"
 - IF **AWAKE**: Ask the patient to eat (candy, juice, low fat milk)
 - IF **UNCONSCIOUS**: Stab with IV D50
- Regular insulin:** ONLY insulin given IV
- NPH:** If mixed, clear-to-cloudy (NPH is cloudy)
- Long-acting:** Do not mix; **NO PEAK**
- Rotate injection sites → do not aspirate/massage
- Always** increase insulin with: (glucose ↑ with any type of stress)
 - Stress
 - Sepsis
 - Sickness
 - Steroids

REMEMBER

TYPE 1: YOU HAVE NONE

- NO insulin being produced
- Patients will need insulin!

TYPE 2: THE PROBLEM IS YOU

- Encourage healthy diet and exercise
- Potential oral medication use
- Insulin (last resort)

INSULIN PUMP

- Give a steady dose of insulin for Type 1 DM
- Check BG 4x/day
- Push bolus at meals

ANTI CONVULSANTS

① PHENYTOIN

ACTION: slows down impulses in the brain that cause seizures

USE: treat and prevent seizures

SIDE EFFECTS: **GINGIVAL HYPERPLASIA!**

▷ Diplopia, Rash, ↓ BP

▷ **THERAPEUTIC RANGE:** 10-20 mcg/mL

KEY: ↓ effectiveness of oral contraceptives

Avoid Milk + antacids (↓ absorption)

Take with food; **Regular dental checkups**

▷ **SIGNS OF TOXICITY:**

Nystagmus, slurred speech

Ataxia, vomiting

② LAMOTRIGINE

ACTION: Inhibits Na⁺ transport → ↓ seizures

USE: Best for partial seizures, mood stabilizer

SIDE EFFECTS: ↑ Risk of suicidal thoughts

Double vision, Headache

Drowsy, Dizzy, N/V

KEY: Watch for worsening depression or mood changes

▷ **DO NOT STOP ABRUPTLY**

▷ ↓ effectiveness for oral contraceptives

③ BARBITUATES (PHENOBARBITAL)

ACTION: stimulates GABA → elevates seizure threshold → ↓ excitement in brain

USE: Tonic clonic, focal, status epilepticus

SIDE EFFECTS: Drowsy, Ataxia, Respiratory depression, ↓ BP

▷ **THERAPEUTIC RANGE:** 15-40 mcg/mL

KEY: Monitor breathing and routine neurochecks

④ VALPROATES (VALPROIC ACID)

ACTION: not fully understood, may ↑ levels of GABA

USE: simple + complex seizures

SIDE EFFECTS: N/V/D, dizzy, drowsy

Tremor, Tinnitus

↑ Bleeding time

↓ Platelet count

KEY: monitor WBC's, platelets, liver enzymes, LFT's

⑤ CARBAMEZEPINE

ACTION: Inhibits Na⁺ transport → ↓ seizures

USE: Tonic clonic, partial, mood stabilizer

SIDE EFFECTS: **AGRANULOCYTOSIS! → infection risk**

Hyponatremia, Dizzy, drowsy, N/V

KEY: Watch for worsening depression or mood change

▷ **DO NOT STOP ABRUPTLY**

ANTIPSYCHOTICS

KEY: For all psychiatric conditions:

DOPAMINE (Think "DOP")

D: Determination for a reward

O: Obsession (addicting)

P: Pleasure

NOREPINEPHRINE (Think "NOR")

N: No hesitation ↑HR ↑BP

O: On Alert

R: Recall is good!

SEROTONIN (Think "SER")

S: Sleep

E: emotion

R: Remember (good memory)

(+) SCHIZOPHRENIA **SX:** experiencing things you **SHOULDN'T** be
(Hallucinations, Delusions, Disorganized Thinking)

(-) SCHIZOPHRENIA **SX:** lack of characteristics that **SHOULD** be there
(Flat/blunt affect, isolation, lack of interest, ↓ speech)

① FIRST GENERATION (AKA TYPICAL/CONVENTIONAL)

SUFFIX: -ZINE + HALOPERIDOL (Haldol)

ACTION: Block D₂ receptors on ALL Pathways of the brain

USE: (+) schizophrenia symptoms

SIDE EFFECTS:

1.) **Extrapyramidal symptoms:** Parkinsons-like **SX**, dystonia, shuffling gait, tardive dyskinesia
KEY: GIVE BENZTROPINE

2.) **Neuroleptic Malignant syndrome:** High Fever, variable BP, ↑HR, muscle rigidity, confusion
KEY: GIVE MUSCLE RELAXANT

3.) Anticholinergic **SX**, orthostatic hypotension, seizures

② SECOND GENERATION (AKA ATYPICAL) more current

SUFFIX: -APINE, -DONE

ACTION: Block D₂ AND serotonin receptors (level out amount of dopamine)

USE: (+) AND (-) symptoms of schizophrenia

SIDE EFFECTS:

1. **Metabolic symptoms:** hyperglycemia (Diabetes), ↑ weight, ↑ cholesterol

2. Anticholinergic **SX**, decreased libido, sedation, orthostatic hypotension

KEY: Clozapine → **AGRANULOCYTOSIS** (watch: immunosuppression)

Risperidone → can be injected q2wks via IM route

MOOD STABILIZERS

USE: For the treatment of BIPOLAR DISORDER

① LITHIUM

ACTION: Helps reduce severity and frequency of manic episodes

‣ unknown how it does this (may inhibit MOA → ↑ activity of the brain)

SIDE EFFECTS: LEUKOCYTOSIS!

increased urination / thirsty
tremors, hypothyroidism
nausea, fatigue

‣ **SIMILAR TO SALT** → **DO NOT** cut out salt from your diet (↑ Risk Toxicity)

THERAPEUTIC RANGE: 0.6-1.2 mEq/L

‣ **WEEKLY** blood draws until levels are steady → then 3mo intervals

‣ **Monitor for salt loss** → Fever, sweating, diarrhea

‣ **Teratogenic**

‣ **1-3 WKS** for full effect

② ANTICONVULSANTS

‣ Valproic Acid → Acute mania attack preferred over lithium due to less side effects

‣ carbamazepine → Agranulocytosis

‣ lamotrigine

③ ANTIPSYCHOTICS

‣ olanzapine

‣ Risperidone

‣ Aripiprazole

④ BENZODIAZEPINES (-PAM, -LAM)

ANTI-ANXIETY MEDS

① BENZODIAZEPINES (-PAM, -LAM)

ACTION: Bind to GABA receptors to enhance the action of GABA

GABA: Neurotransmitter that slows (calms) the speed of brain activity

SIDE EFFECTS: sedation, ↓RR, ↓BP, ↓HR, amnesia

⚠️ **KEY:** **FAST ACTING** (Think: Pam is riding in a Benz, which is fast) →
↓
short term use only! can cause **DEPENDENCY** Antidote: Flumazenil ⚠️

② SSRI'S (SELECTIVE SEROTONIN REUPTAKE INHIBITORS)

ACTION: see Antidepressants

USE: Anxiety, depression, PTSD, OCD (Think: Teens (-tine) have a lot of anxiety)

SIDE EFFECTS: weight gain, insomnia, sexual dysfunction

4WKS for full effect!

⚠️ **KEY:** Do not take with St. John's Wort

SUFFIX: -TINE, -LINE, -PRAM

③ ATYPICALS - BUSPIRONE

USE: Anxiety, PTSD, OCD, Panic

1-2WKS for full effect!

SIDE EFFECTS: Dizzy, headache, nausea

⚠️ **KEY:** Does **NOT** cause sedation **OR** dependency!
Take with **food** if GI upset

④ OTHERS: TRICYCLIC ANTIDEPRESSANTS

BETA BLOCKERS

BARBITUATES (-barbital) = stays in system for
3-5 days

ANTIDEPRESSANTS

KEY: Take 2-4 WKS to take full effect

THINK: 4 letters in "SSRI" → 4 WKS

> RISK FOR SUICIDE

① SSRI'S (SELECTIVE SEROTONIN REUPTAKE INHIBITORS)

ACTION: Inhibit the reuptake of serotonin back into the neuron → more serotonin floating around in the brain → improve mood

SIDE EFFECTS: weight gain, insomnia, sexual dysfunction

RISK: serotonin syndrome → Agitation, hallucinations, fever, tremors, sweaty

SUFFIX: -TINE, -LINE, -PRAM

② SNRI'S (SEROTONIN - NOREPINEPHRINE REUPTAKE INHIBITORS)

ACTION: Inhibit the reuptake of serotonin AND norepinephrine into the neuron → more floating in the brain

SIDE EFFECTS: loss of appetite, insomnia, sexual dysfunction

③ TCA'S (TRICYCLIC ANTIDEPRESSANTS) -TRYPTLINE -IPRAMINE

ACTION: TRI = 3 actions → 1. Inhibit serotonin reuptake

USE: depression

anxiety

insomnia

fibromyalgia

2. Inhibit norepinephrine reuptake

3. Inhibit Alpha receptors → orthostatic hypotension

Histamine → sedation

Muscarinic receptors → anticholinergic sx

SIDE EFFECTS: All of these
+ urinary retention

KEY: Be careful in the sun ("cycle") → ↑ fiber / ↑ fluids

④ MAOI'S (MONOAMINE OXIDASE INHIBITORS)

Phenelzine
Ex: Isocarboxazid

ACTION: Blocks the destruction of serotonin, norepinephrine, dopamine and TYRAMINE → increase ALL of these in the brain

USE: Depression, panic disorder

SIDE EFFECTS: Agitation, anxiety, ↓ BP, HYPERTENSIVE CRISIS (interacts with many meds) DO NOT EAT TYRAMINE FOODS!! (chocolate, avo, wine, cheese)

⑤ ATYPICALS

A. Bupropion (Wellbutrin) → for depression + smoking cessation

B. Trazadone → for depression (high dose) + insomnia

ANTIPARKINSON MEDS

REMEMBER: Patients have either **TOO LITTLE** Dopamine or **TOO MUCH** Acetylcholine

Characteristics:

- Pin Rolling Tremors
- Blank expression
- Stooped Posture
- Shuffling gait
- Muscle Rigidity
- Possible mental deterioration

① ANTICHOLINERGIC AGENTS (BENZTROPINE)

ACTION: Antagonizes Acetylcholine and histamine receptors → ↓ Ach

USE: Decrease muscle rigidity and tremors

SIDE EFFECTS: Anticholinergic symptoms: Dry mouth, dry eyes, blurry vision, urinary retention, constipation

TEACH: chew gum (dry mouth)
wear sunglasses, ↑ fiber, ↑ fluid

KEY: Do **not** give to a patient with glaucoma; **No alcohol**; **Don't stop abruptly**

② MAO-B INHIBITORS (RASAGILINE aka AZILECT)

ACTION: Increase dopamine by inhibiting MAO activity

USE: Improve shaking + stiffness, Decrease amount of offtime (periods of

SIDE EFFECTS: Dizzy, drowsy, nausea, heartburn, dry mouth (slow movement)

KEY: **Avoid foods with Tyramine!** → cheese, chocolate, wine, avo, beer

③ COMT INHIBITORS (-CAPONE)

ACTION: Decrease wearing off of carbidopa / levodopa by blocking COMT enzyme

USE: With carbidopa / levodopa (slows breakdown of levodopa)

④ DOPAMINE AGONISTS:

CARBIDOPA / LEVODOPA → decrease muscle rigidity and tremors

ACTION: Increase Dopamine → carbidopa prevents levodopa from being converted into Dopamine in the bloodstream → ↑ Dopamine entering the brain

SIDE EFFECTS: N/V, drowsy, dyskinesia, Dark urine / sweat, ↓ BP

KEY: May take ~3WKS to be effective, ↑ protein = ↓ effectiveness

NO MAOI'S ⚠️

ROPINIROLE (AKA REQUIP) → improve tremors, muscle spasm, stiffness

May also be used for Restless leg syndrome

SIDE EFFECTS: Drowsy, confusion, orthostatic Hypotension → **FALL RISK** ⚠️

SPASMOLYTICS

MUSCLE RELAXANTS

① DANTROLENE → Direct acting on the muscle cell

USED FOR: ↓ muscle spasticity in spinal injuries, cerebral palsy, multiple sclerosis, malignant hypothermia

ACTION: Prevents calcium release in skeletal muscle = ↓ muscle spasticity

SIDE EFFECTS: Drowsy, muscle weakness, GI upset
↳ Hepatic toxicity

WATCH: Liver enzymes

② BAGLOFEN → "used if you pull your BACK"

Centrally Acting on the CNS

USED FOR: same as Dantrolene!

ACTION: Enhances GABA in the CNS = ↓ muscle spasticity

Σ What is GABA? A neurotransmitter in the brain that slows the speed of messages between cells

SIDE EFFECTS: Drowsy, nausea, GI upset, constipation

TEACH: ↑ Fiber and ↑ fluids to combat constipation

③ TIZANIDINE

Centrally Acting on Alpha₂ adrenergic agonist

USED FOR: same as Dantrolene and Baclofen!

ACTION: Inhibits presynaptic motor neurons to decrease # of spasms, alleviate clonus and improve mobility

SIDE EFFECTS: LESS muscle weakness than Baclofen

↳ Dry mouth, drowsy, dizzy

ADHD MEDS

① METHYLPHENIDATE (RITALIN)

> short Acting

② AMPHETAMINE (ADDERALL)

> short Acting: onset 30-45 min
wear off 3-6 HRS

> long Acting: ("XR" = extended release)
onset 45-1HR
wear off 8-12 HRS

> KEY: Do not take at night - insomnia
Take immediately before/after a meal
Monitor weight

SIDE EFFECTS:

- > weight loss
- > decreased appetite
- > insomnia
- > dysrhythmias
- > moody / irritable

WITHDRAWAL MEDS

① ALCOHOL → as early as 6 hours KNOW TIME OF LAST DRINK! PEAK: 24-48 HRS

- A. Benzodiazepines: ↓ withdrawal, ↓ Risk seizures, stabilize vitals
- B. Anticonvulsants: ↓ Risk seizures
- C. Antihypertensives (clonidine) + Beta Blocker: ↓ HR, ↓ BP, ↓ craving

② OPIOID

A. Methadone (long term maintenance)

③ NICOTINE

- A. Bupropion
- B. Nicotine Replacement
- C. Varenicline

④ ABSTINENCE - suppress patient craving

- A. Disulfiram
- B. Naltrexone
- C. Acamprosate → ↓ anxiety / Restlessness
"A camp makes you happy"

BAD with alcohol:

- > N/V, sweating
- > Palpitations

ALCOHOL ADDICTION TREATMENT

M E D I C A T I O N S

NALTREXONE → REDUCES OR ELIMINATES ALCOHOL CRAVING

ACAMPROSATE → HELPS PATIENT ABSTAIN FROM ALCOHOL

TOPIRAMATE → DECREASE ALCOHOL CRAVINGS

DISULFIRAM → REACTION WITH ALCOHOL CAUSES UNPLEASANT PHYSICAL EFFECTS

NALTREXONE → HELP REDUCE ALCOHOL CRAVING

ALCOHOL WITHDRAWAL MEDICATIONS

R I S K O F S E I Z U R E S

seizures MAY occur 24-48 hours after last drink

→ we want to stay prepared for this risk by giving:

BENZODIAZEPINES

ANTICONVULSANTS

BETA BLOCKERS

MAGNESIUM SULFATE

THIAMINE (B1), FOLIC ACID

WITHDRAWAL SYMPTOMS: tremors, vomiting, anxiety, irritability
(as early as 6 hours) high blood pressure, ↑HR, confusion

SEIZURE SYMPTOMS: extreme confusion + agitation, ↑HR, fever, sweating, LOC, aura, headache

NSAID'S + ACETAMINOPHEN

TYPES:

1. Naproxen
2. salicylate Acid (Aspirin)
3. Acetylsalicylate Acid (also Aspirin)
4. Ibuprofen
5. Ketorolac (Toradol)

AVOID WITH:

- > Vitamin E
- > Garlic
- > Ginkgo
- > Omega 3 oils

CHARACTERISTICS:

KEY: Does not cause CNS symptoms

- > **ANTIPYRETIC** → decrease fever
- > **ANTI-INFLAMMATORY** → decrease prostaglandin synthesis
- > **ANALGESIC** → decrease pain
- > **ANTI-PLATELET** (Aspirin) → **RISK = Bleeding**

NURSING CONSIDERATIONS:

- > NSAIDs affect the **kidneys!**, liver, heart and GI system
- > **DO NOT use** in patients with
 - > Asthma → bronchospasm
 - > Ulcers → worsen + GI bleed
 - > Kidney disease = ↑ Cr, ↑ BUN, Renal injury in long term
 - > Ibuprofen = ↑ **RISK** HTN + stroke
 - > Heart failure / Cardio disease (most contain Na⁺ = ↑ edema)
- > can **increase risk** of blood clots except Aspirin
- > **OK** to take if on **PPI** (proton pump inhibitor)
- > **Take with food!** **Lowest Dose!** **NO Alcohol!**

ASPIRIN KEY THINGS:

1. Has all the same properties **AND** **thins blood**
2. **Monitor** for s/s of bleeding, hypotension, tachycardia
3. **Monitor** for **toxicity** = earliest sign = **TINNITUS**
4. **DO NOT GIVE** to Kids → Reye's syndrome!

NOT AN NSAID: TYLENOL (Acetaminophen)

1. Used instead of NSAIDs to ↓ **RISK** of bleeding in PVD + hemophilia
 2. Analgesic, Antipyretic
NOT anti-inflammatory
 3. can be taken with Ibuprofen
 4. **TOO MUCH = liver damage** = monitor LFT's
- ↳ Antidote: **ACETYL CYSTEINE**

ANTIBIOTICS

BROAD SPECTRUM (-CILLIN / -CEF / -CEPH)

- > safe for pregnancy + breastfeeding
- > ALLERGY to -cillins, ASSESS type of reaction FIRST before giving a -ceph drug

-CILLINS

1. Watch with oral contraceptives
2. Take with food! shake well!
3. SIDE EFFECT: bleeding (watch clot time)

-CEF / -CEPH

1. can cause C diff; Does not treat C diff
2. Take with food if GI upset
3. SIDE EFFECT: diarrhea (common)

-THROMYCINS (AZITHROMYCIN / ERYTHROMYCIN)

BIG LIVER TOXIC → watch AST/ALT and when a patient is taking Tylenol

- > "Throw" the QT interval → ↑QT interval = cardiac arrest
- > Take on empty stomach with full glass of water → NO FOOD

-CYCLINES (TETRACYCLINE / DOXYCYCLINE)

- > Teratogenic! → also watch oral contraceptives
- > Tooth discoloration
- > AVOID the SUN → "cycling in the sun"

WHEN TAKING:

- > NO food → Empty stomach with water
- > NO Ca²⁺ + Iron → blocks absorption
- > sit up 30min after taking (Risk: Esophagitis)

METRONIDAZOLE (-Dazole) AKA FLAGYL

USE: C diff, STI's (Trichomoniasis)

AVOID: Alcohol during + 3 days after treatment

NML: Brown urine, metallic taste

BAD: new rash / skin peeling = Stevens Johnson syndrome

ANTIFUNGAL (FLUCONAZOLE)

USE: candidiasis

Watch in pregnancy: teratogenic, only given when benefit outweigh risk

SIDE EFFECT: GI liver toxicity = watch LFT's, CBC, Renal

CONTRAINDICATED: Phenytoin, Digoxin, anticoags

HIGHLY TOXIC ABX (-mycins / -micin)

USE: MRSA + C diff

"A SIN TO GIVE"

- > TOXIC to the kidneys, Ears + Liver

1. Monitor Peak/Trough level 15-30min BEFORE given

2. Monitor S/S Toxicity:

Kidneys: ↑BUN, ↑Cr, ↓UOP

Ears: vertigo, tinnitus

3. very strong! Monitor IV site q 30 min!!

Ⓐ Glycopeptide = vancomycin

> Red Man syndrome → NOT deadly = sudden onset rash, flushing, hypotension
→ slow infusion over 1hr

Ⓑ Aminoglycosides = Gentamicin, Tobramycin

> very TOXIC w/ vanco > NO Red Man syndrome

UTI MEDS (SULFA- / -FLOXACIN)

Ⓐ SULFA - stops folic acid synthesis

> Teratogenic

> Avoid the sun → "sit on sofa inside"

> Watch for ↑ crystals, ↑ specific gravity

> Encourage water + Folic acid

Ⓑ FLOXACIN

> Avoid the sun

> Take with water → NO FOOD

> Report muscle pain (↑ Risk Achilles tendon tear)

PHENAZOPYRIDINE (Pyridium)

NOT an ABX → supportive care

Normal: Red orange body fluid!

Watch: Jaundice (Big liver toxic)

KEY TAKEAWAY:

1. Ineffective oral contraceptive: -cillins / -cyclins

2. NO FOOD, WITH WATER

-Thromycin
-Tetracycline
-Floxacin
"Move The Food"

3. NO SUN: -cycline (cycling)

sofa (sofa)

-floxacin

4. Extremely TOXIC: MYCIN

5. Teratogenic: -cycline

sofa
antifungal

FINISH
THE
PRESCRIPTION

ANTIVIRAL MEDS

WHAT ARE THEY? Agents used to treat the diseases caused by viruses like **chickenpox** and the **common cold**

① ACYCLOVIR

USED FOR: Herpes simplex virus →

{ **Drug of choice for KIDS**

> Given IV, Oral, Topical

WATCH: signs of Renal impairment

> pregnancy

> ↑ doses = **Nephrotoxicity** and **Neurotoxicity** (headache, confusion, hallucinations)

② GANCICLOVIR

USED FOR: Cytomegalovirus → infections affecting the eyes, respiratory tract and liver

> Given IV Route

WATCH: Bone marrow suppression (big side effect)

> Monitor signs of infection and CBC

FOR BOTH:

1. Prevent viral DNA replication
2. Do **NOT** cure → only reduce symptoms
3. **Avoid sex** when lesions are present (use condoms)
4. **Topical** → wear gloves when administering
5. **IV** → watch for phlebitis (common)
6. May cause **GI upset** and **nephrotoxicity!**

⚡ **MODE OF ACTION:**
-vir drugs **inhibit** viral DNA polymerase and prevent replication of viral DNA

OPIOID ANALGESICS

ACTION: Bind to opioid receptors in the central nervous system to sedate the body

CHARACTERISTICS:

- > Analgesic
- > sedative
- > Low and slow vitals ($\downarrow$ RR, $\downarrow$ HR, $\downarrow$ BP)

DOES NOT decrease fever or inflammation

NURSING CARE:

- > Teach breathing exercises
- > Always **monitor** vitals / orthostatic hypotension
- > Dizzy? sit the patient down **FIRST**, then assess
- > Antidote: Naloxone

IV ADMINISTRATION:

1. 2-3 min IV push (Don't bolus = DEATH)

2. Reassess 15-30 minutes

(PO opioids = reassess in 1hr)

3. Watch for **Risk of overdose!** Also: respiratory depression, elderly, underlying disease (COPD/Asthma), post op

4. **NORMAL:** IV Burn ✓

itching/N/V → goes away ✓

slow GI → give stool softener, fluids, ambulate ✓

SIGNS OF OVERDOSE:

- > $\downarrow$ RR < 12 , unarousable, dizzy, confused
- > easily falls asleep when talking
- > cold, clammy skin, deep gurgling

ANTIDOTE: NALOXONE (Narcan)

KEY: 1-2hr $\frac{1}{2}$ life → reassess and give another dose if needed

ASSESS: ABC's, O₂, Tell HCP, 2nd Dose

TYPES:

1. Morphine
2. Hydromorphone
3. codeine
4. Oxycodone
5. Oxycontin
6. Fentanyl

PCA PUMP POST OP:

- > client controlled **ONLY**
- > Notify HCP to **↑ dose** if:
- > Twice the dose of med given and still reports pain

FENTANYL (for chronic pain)

- > can give if allergy to codeine
- > **SE:** constipation
- > **common:** Patch (**not** give immediate relief) → 17hr full effect
- Remove old patch before new!!

TUBERCULOSIS MEDS

KEY THINGS:

1. Treatment lasts approximately 6-9 MONTHS
2. sputum samples are needed q2-4 WEEKS
3. NO LONGER INFECTIOUS WHEN: 3 Negative sputum cultures on 3 different days
4. Always wear N-95 mask
5. Family should be tested (The baby is safe in pregnancy :))

① RIFAMPIN

- > causes Red/orange tears, sweat, urine = NORMAL
 - Teach to wear glasses, NOT contacts
- > Makes oral contraceptives ineffective
- > Monitor jaundice → LIVER TOXIC

② INH = ISONIAZID

- > interferes with absorption of vitamin B6 (causes peripheral neuropathy)
 - Take B6 25-50mg/day
- > causes neuropathy → report numbness, tingling, ataxia
- > LIVER TOXIC → monitor jaundice, dark urine, fatigue, elevated LFT's

③ ETHAMBUTOL

- > "E" for EYES → report blurred vision, vision change, color change
- > Get routine eye exams
- > Generally safe for liver

④ OTHER: PYRAZINAMIDE (Hepatotoxic!)

CHEMOTHERAPY MEDS

① LEUPROLIDE (Prostate cancer)

ACTION: Decrease levels of Testosterone

SIDE EFFECTS: hot flashes, bone loss, ↓ libido, gynecomastia (↑ breast tissue)

KEY: Encourage **Ca⁺** and **Vit. D**, assess for bone pain / urine output / bladder distention
➤ **Monitor** prostate specific antigen + testosterone level

② TAMOXIFEN (Breast cancer)

ACTION: competes with estrogen for binding sites in the breast + stops the growth of estrogen dependent cancer

SIDE EFFECTS: hot flashes, GI upset, ↑ Ca⁺, ↑ Risk PE, ↑ Risk endometrial cancer

KEY: **Monitor** calcium levels, CBC, Platelets

{ **Think:** a lady named **Tami** for **Tamoxifen** }

③ INTERFERON ALPHA-2B (Biologic Response Modifier)

USE: cancer + hepatitis → "interferes" with cancer

ACTION: ↑ immune response to tumors + viruses; ↓ cancer cell production

SIDE EFFECTS: flu-like sx, bone marrow suppression, ♥ toxicity, neurotoxicity, GI upset, Alopecia

④ DOXORUBICIN (Antitumor antibiotic)

ACTION: binds to DNA → inhibits DNA/RNA synthesis → death to rapidly dividing cells

SIDE EFFECTS: GI upset, Alopecia, ♥ toxicity
➤ Bone marrow suppression = anemia, neutropenia, thrombocytopenia
➤ Red discoloration of urine, sweat, tears

KEY: **Monitor** for s/s of infection

➤ Give antiemetics

➤ **Monitor** ♥ function and CBC

{ **Think:** "Ruby Red" for red discoloration, the heart and blood for bone marrow ☺ }

CHEMOTHERAPY MEDS

⑤ FOR A VARIETY OF CANCERS:

VINCRIStINE (Anti-mitotic)

ACTION: stops cell division in mitosis

SIDE EFFECTS: Peripheral neuropathy, Alopecia
‣ Phebitis, GI upset

KEY: Monitor s/s of neuropathy (tingling)

- Give antiemetics
- Administered best by PICC line or central line
- Does **NOT** cause bone marrow suppression

Think: "VIN" is wine in French → Too much wine you can't feel your arms and legs

CYCLOPHOSPHAMIDE (Alkylating Agent)

ACTION: Inhibits protein synthesis

SIDE EFFECTS: GI upset, alopecia

‣ **BONE MARROW SUPPRESSION** = anemia, leukopenia, thrombocytopenia
‣ **Hemorrhagic cystitis!!**

KEY: increase fluid intake

- **monitor** for blood in urine
- **Monitor** CBC, give anti emetics

TRAUMA MEDS

LIDOCAINE:

ACTION: temporarily numbing and relieving pain

- > IV = to treat VFib / VTach
- suppresses ventricular cells
- ↓ diastolic depolarization

USE FOR: Local Anesthetic (and topical)
Ventricular Arrhythmias

EPINEPHRINE:

ACTION: stimulate alpha + beta adrenergic receptors
= relaxes ♥ + lung smooth muscle + dilation of vessels in skeletal muscle

USE FOR: bronchodilation in these cases:

1. anaphylaxis
 2. hypersensitivity reaction
 3. acute asthma attack
- chronic simple glaucoma
- } Bronchoconstricts = BAD

ATROPINE:

ACTION: inhibits ACh in lung smooth muscle + ♥ muscle
blocks SA node + AV node = ↑ HR

USE FOR: decreasing respiratory secretions
sinus bradycardia

WATCH FOR: overdose of anticholinergic drugs

NARCAN:

ACTION: treat overdose of narcotic med

USE FOR: opioid induced toxicity
Respiratory depression

Neonates affected by narcotics given during labor

"WE ALL NEED SOMEBODY TO LEAN ON"

ISO, HYPO, AND HYPERTONIC FLUIDS

GOAL: Replace lost fluids from dehydration, shock, hemorrhage, burns, etc.

REMEMBER: The **WATER** goes where the **SALT** flows

ISOTONIC EQUAL concentration of SOLVENT and SOLUTE TISSUE CELLS →  TYPES: 0.9% Normal saline 5% Dextrose in water (D5W) 5% Dextrose in .225% saline - Lactated Ringers → Good for prego moms and burns KEY: D5W turns into a hypo tonic solution (absorbs dextrose) USE: ↑ extracellular fluid > Blood loss, Dehydration surgery, Burns NURSES: 1. D5W is isotonic in the Bag, hypotonic in the body 2. LR → Burns + Preggo Mom - NOT liver impaired - NOT hyperkalemic	HYPOTONIC (LOW) ↓ concentration of SOLUTE in solution  SWELL TYPES: 0.45% NS ($\frac{1}{2}$ NS) 0.225% saline ($\frac{1}{4}$ NS) 0.33% saline ($\frac{1}{3}$ NS) KEY: watch for cell lysis → TOO BIG they will burst watch for depletion of circulatory system USE: Rehydrate cells > Dehydration > DKA/HHS > NOT for ↑ICP, burns or Trauma! NURSES: 1. Give SLOWLY to prevent cellular edema 2. Risk of Hypovolemia Tachy, ↓BP, cellular edema/damage 	HYPERTONIC (HIGH) ↑ concentration of SOLUTE in solution (SALTY)  SHRINK TYPES: > 5% Dextrose in 3% saline 0.9% NS 5% saline > 5% Dextrose in 10% Dextrose 0.45% saline > 5% Dextrose in LR KEY: very hard on the vein very concentrated → give via central line USE: Hypovolemia - Hyponatremia - cerebral edema TOO MUCH = PULMONARY EDEMA NURSES: 1. Give slowly (hard on vein) 2. Watch fluid overload bounding pulses, JVD crackles, edema
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KEY: Dextrose is sugar!

> DO NOT give to hyperglycemic patients

LACTATED RINGERS has potassium!

> DO NOT give to liver impaired or hyperkalemia

COMMON MEDICATION ANTIDOTES



ACETAMINOPHEN → N-ACETYLCYSTEINE (MUCOMYST)

ANTICHOLINERGICS → PHYSOSTIGMINE

ASPIRIN → SODIUM BICARBONATE

BENZODIAZEPINE → FLUMAZENIL

MAGNESIUM SULFATE → CALCIUM GLUCONATE

CARBON MONOXIDE POISON → OXYGEN

COUMADIN / WARFARIN → VITAMIN K

DIGOXIN → DIGIBIND OR DIGIFAB

HEPARIN → PROTAMINE SULFATE

INSULIN REACTION → GLUCAGON

OPIOID / NARCOTICS → NALOXONE (NARCAN)

POISONING → ACTIVATED CHARCOAL

IRON → DEFEROXAMINE